An agent-based model of conspicuous consumption and cultural change

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Introduction

- Meeting climate goals means a reduction in global greenhouse gas emissions
- Decrease in demand-side emissions requires radical changes in lifestyles
- Individual pro- or anti-environmental behavioural choices form a constituent part of lifestyles
- Speed of changes and rate adoption of pro-environmental behaviours can be modulated by culture
- Generate awareness of what roadblocks may appear due to interactions between culture and behavioural diffusion

Research questions

- How do pro-environmental **behavioural diffusion** processes interact with **longer-term** cultural evolution, when considering the immediacy of climate change?
- How can the increasing popularity of one pro-environmental behaviour **spur on or inhibit** the diffusion of others?

Considerations and literature drawn on

- Culture and cultural evolution
  - Culture as socially transmitted information - What social influence influences (Axelrod, 1997)
  - Biased social learning (Boyd and Richerson, 1988)
  - Self-defining identity (Barth, 1969; Grimson, 2010; Smaldino, 2019)
- Behavioural diffusion
  - Opinion dynamics (DeGroot, 1974; Hegselmann, Krause, et al., 2002)
  - Network structures (Konc and Savin, 2019)

Model structure - ABM with social network

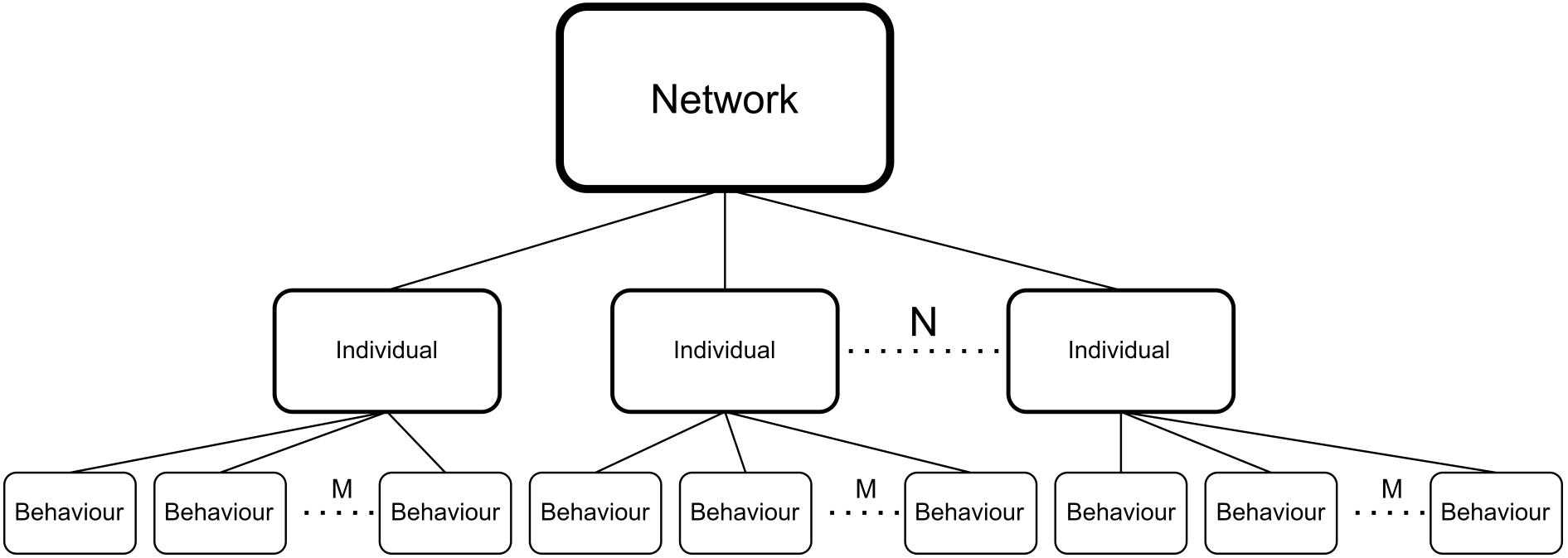


Figure 1: Individuals in a small world social network communicate regarding behaviours

- Individuals $n = \{1, ..., N\}$, each have multiple continuous environmentally related behaviours $m = \{1, ..., M\}$ that evolve over discrete time $t = \{0, ..., T\}$

Model structure - Components

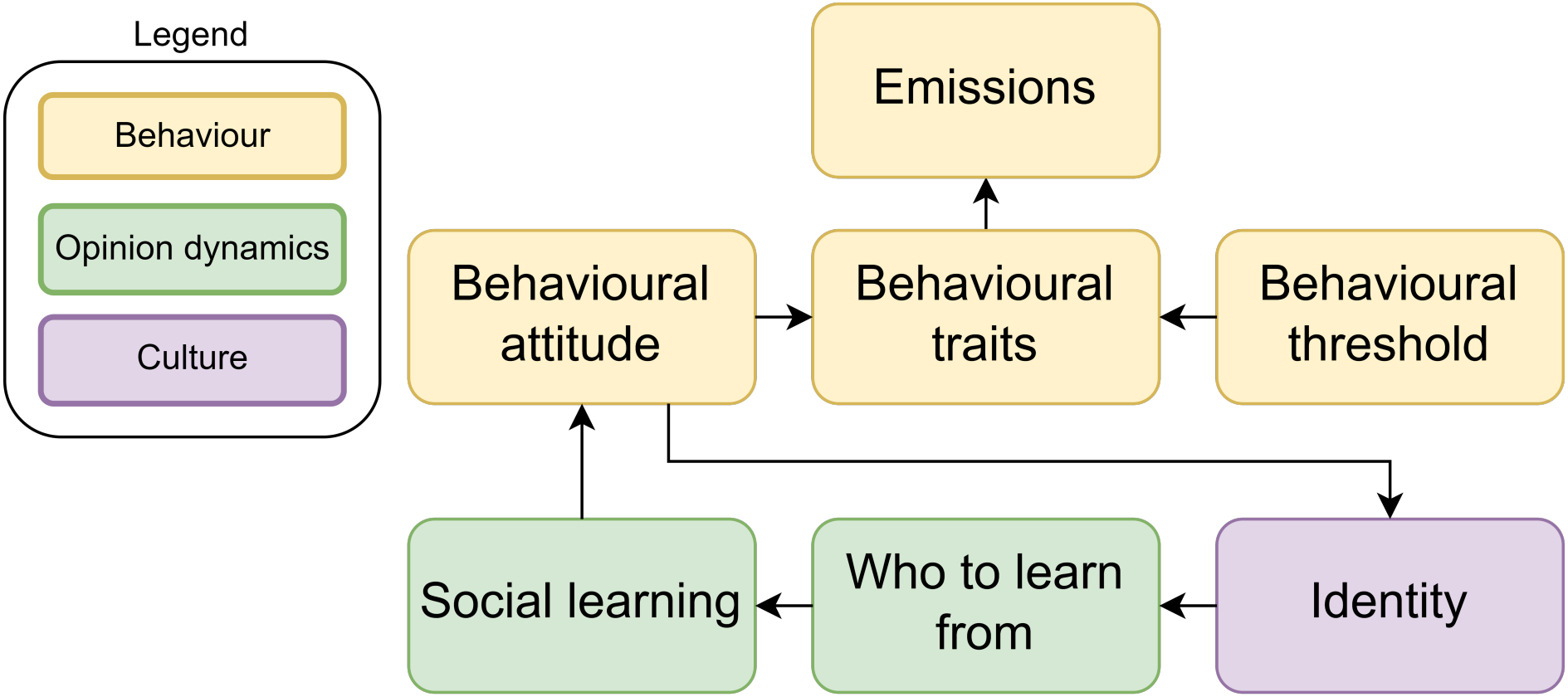


Figure 2: Model components, arrows indict flow of influence

Model structure - Performing behaviours

- Favourable attitudes towards a behavioural option are not sufficient, one must also have the means or control to perform them (Ajzen, 1991)
- For each behaviour m, the continuous trait $B_{t,n,m} \in [-1, 1]$ derived from the attitude $A_{t,n,m} \in [0, 1]$ and threshold $T_{t,n,m} \in [0, 1]$

$$B_{t,n,m} = A_{t,n,m} - T_{t,n,m} \tag{1}$$

- Values for $T_{0,n,m}$ and $A_{0,n,m}$ are generated separately using a Beta distribution
- $A_{t,n,m}$ is dynamic but $T_{t,n,m}$ fixed
- Behavioural trait values decide emissions $e_{n,m}$, assume green behaviours produce none

$$e_{n,m} = \begin{cases} 0, & \text{if } B > 0 \\ 1, & \text{otherwise} \end{cases} \tag{2}$$

Model structure - Inter-behaviour dynamics

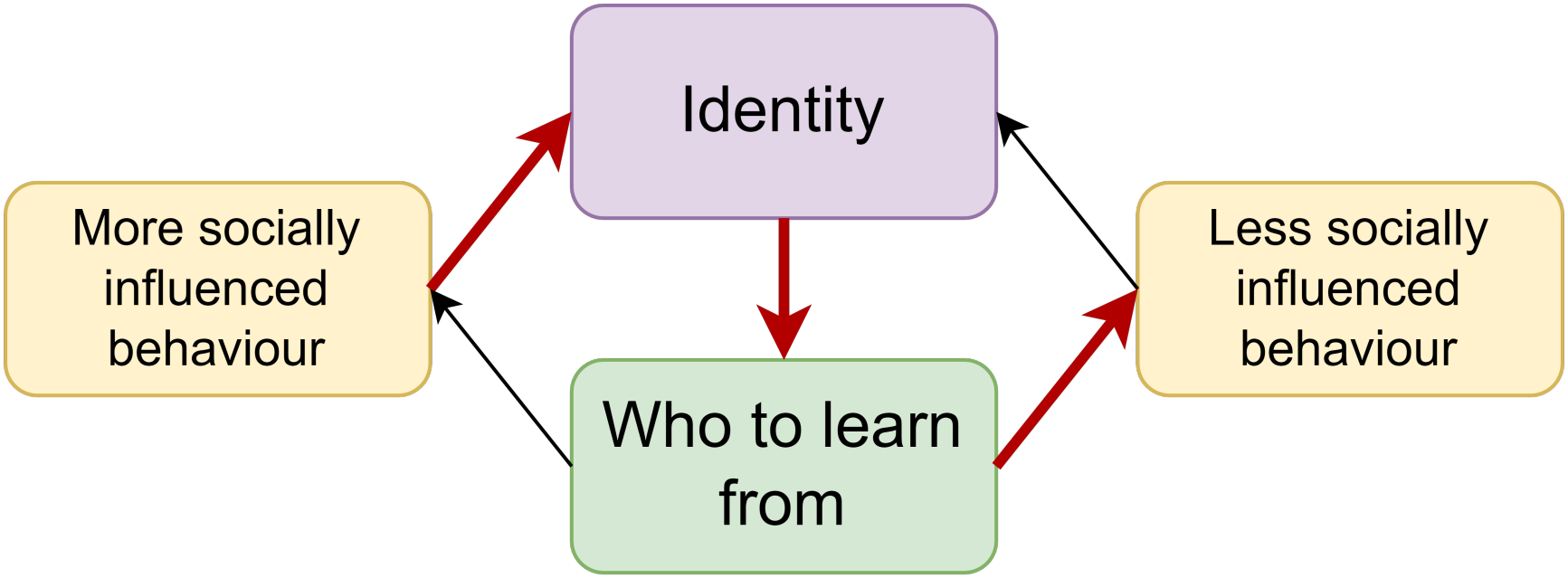


Figure 3: Indirect influence of one behaviour on another through agent identity

- Social interactions, $S_{t,n,m} \in [0,1]$, play a greater role in attitudes towards behaviours that contribute to conspicuous consumption, $\phi_m \in [0,1]$

$$A_{t+1,n,m} = [1 - \phi_m]A_{t,n,m} + [\phi_m]S_{t,n,m} \tag{3}$$

Model structure - Imperfect social learning

- Individuals aggregate the influence $A_{t,k,m}$ of all $k = \{1, ..., K_n\}$ neighbours in their ego network, weighted by their identity similarity, $\alpha_{n,k} \in [0, 1]$
- There is imperfect social learning represented as a gaussian error, ϵ

$$S_{t,n,m} = \sum_{k=1}^{K_n} \alpha_{n,k} A_{t,k,m} + \epsilon \tag{4}$$

Model structure - Network weighting matrix

- Individuals that identify with a particular group tend to assimilate their norms and they distance themselves from the out-group
- This phenomenon is driven by the distance in agent-neighbour identities $|I_n - I_k|$ and modulated by confirmation bias θ

$$\alpha_{n,k} = \left[\frac{1}{\sum_{k=1}^{K} e^{-\theta|I_n - I_k|}} \right] e^{-\theta|I_n - I_k|} \tag{5}$$

Model structure - Representing identity

- Agent identity is defined through their average opinion or attitude, $\bar{A}_{t,n}$, towards M total environmental behaviours
- This average attitude is weighted by recent history with diminishing importance, $D_s = \{1, \beta\delta^1, \beta\delta^2, ..., \beta\delta^{T_\rho}\}$ up to T_ρ time into the past
- Where $\beta \in [0, 1]$ is the present bias and $\delta \in [0, 1]$ the discount parameter

$$I_{t,n} = \left[\frac{1}{\sum_{s=0}^{T_\rho} D_s} \right] \sum_{s=0}^{T_\rho} D_s \bar{A}_{t-s,n} \quad (6)$$

$$\bar{A}_{t,n} = \frac{1}{M} \sum_{m=1}^{M} A_{t,n,m} \quad (8)$$

$$D_s = \begin{cases} 1, & \text{if } s = 0 \\ \beta\delta^s, & \text{if } s \geq 1 \end{cases} \quad (7)$$

Model structure - Representing identity

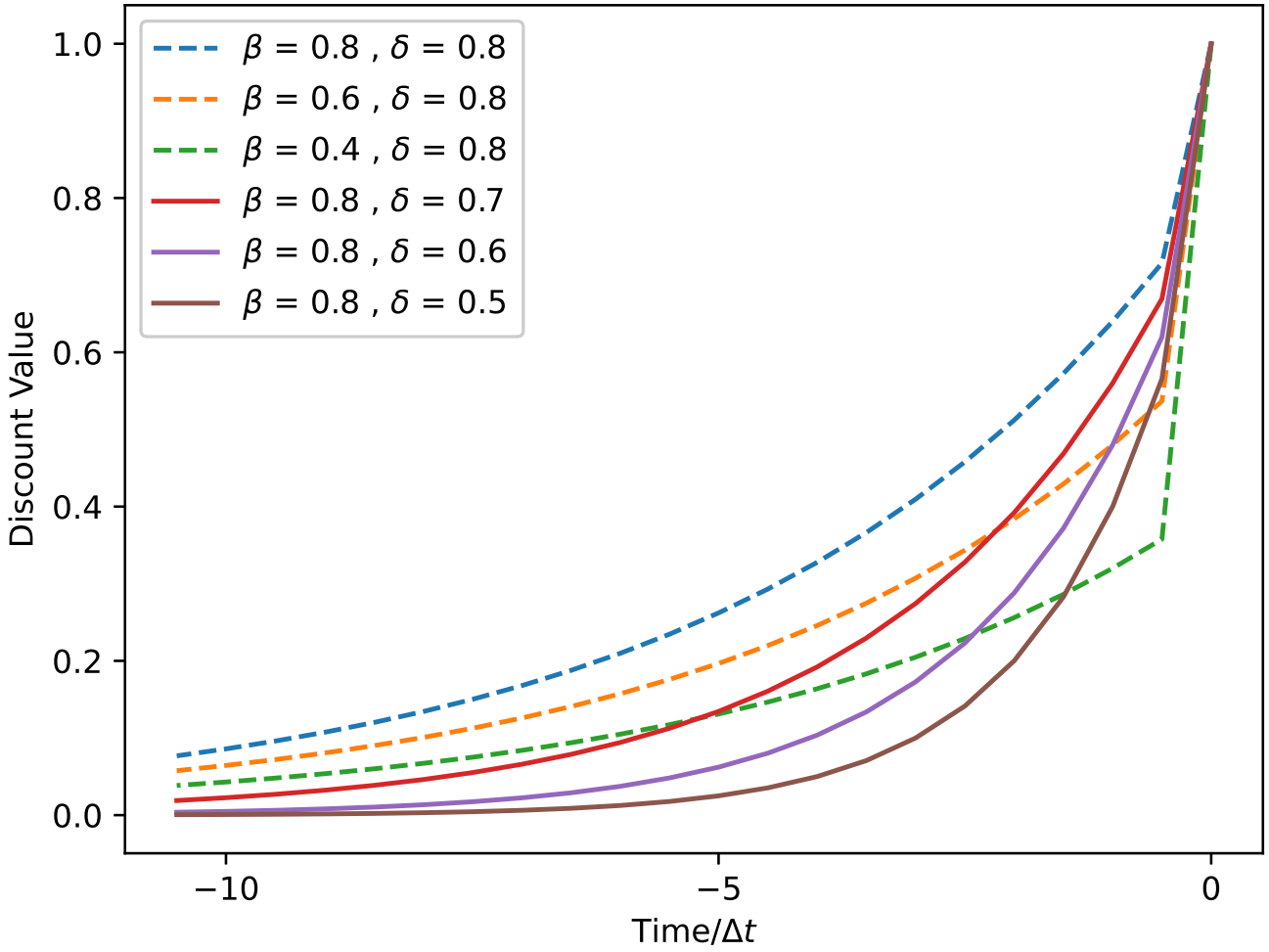


Figure 4: Discount Factor D_t

Results - Initial conditions dominate

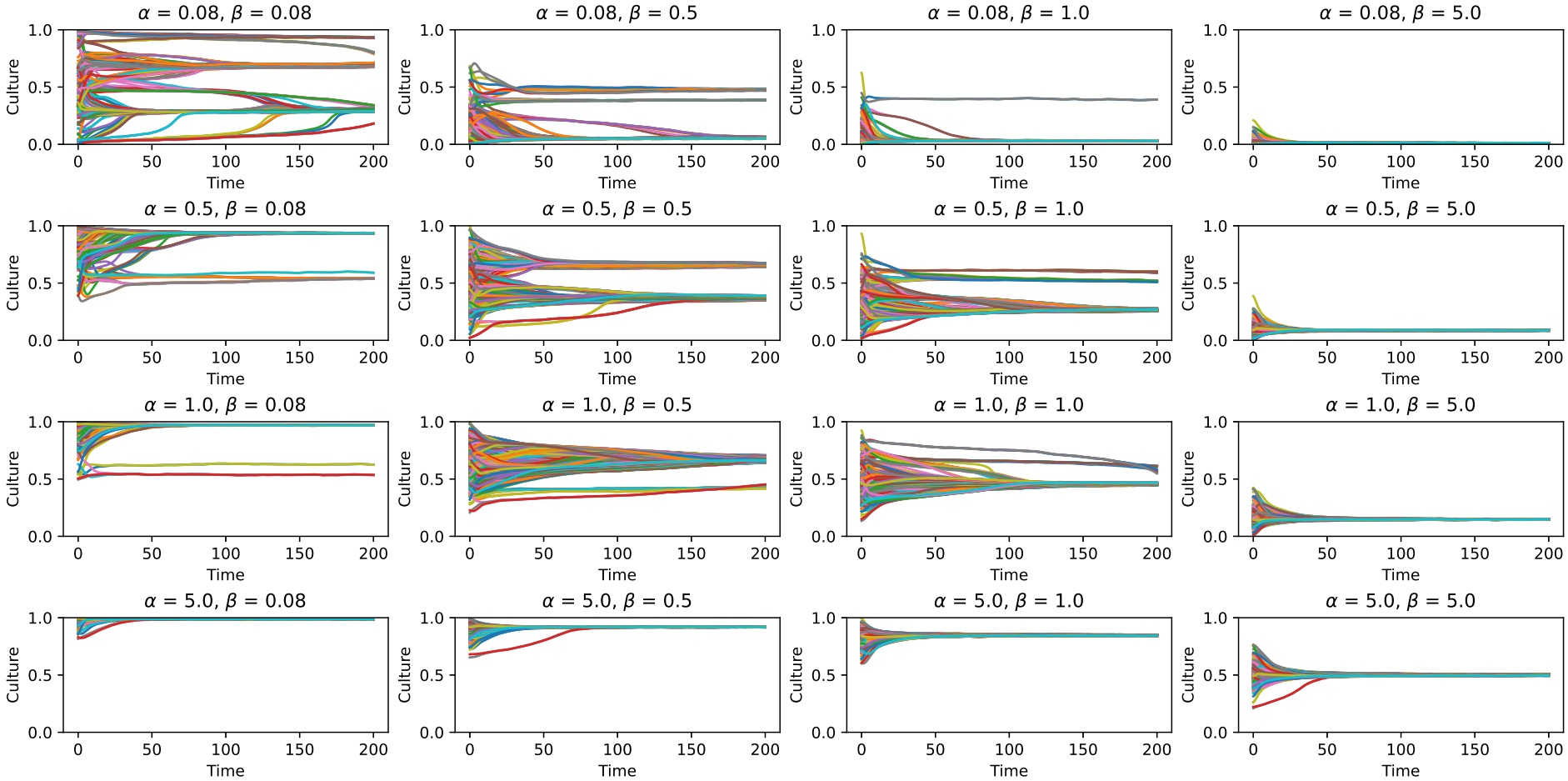


Figure 5: Varying initial attitude beta distributions lead to radically different identity outcomes. Backed by Sobol sensitivity analysis with 30,000 experiments

Results - Consensus formation

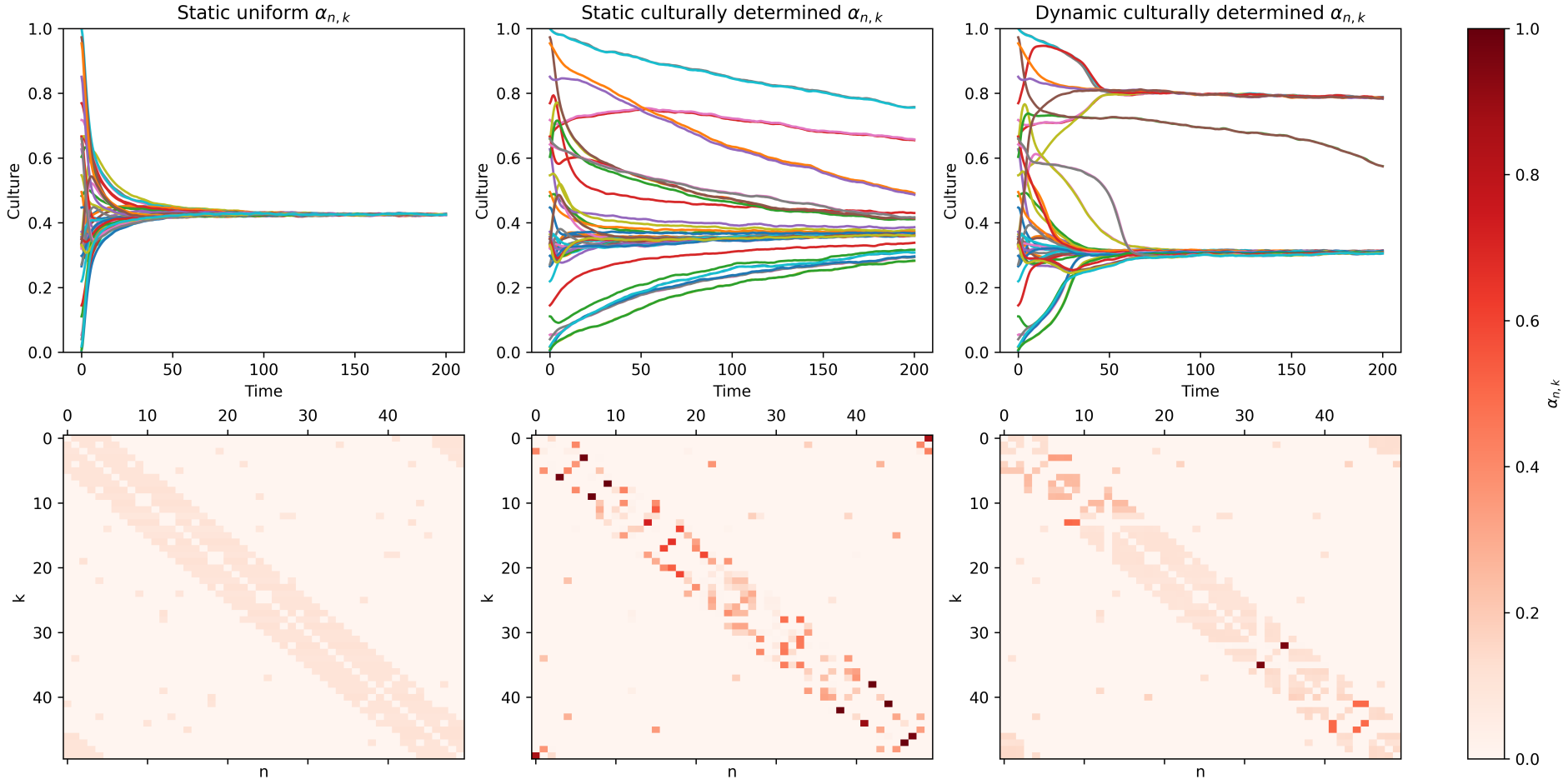


Figure 6: Agent identity over time and final network link strength

Results - Identity homophily

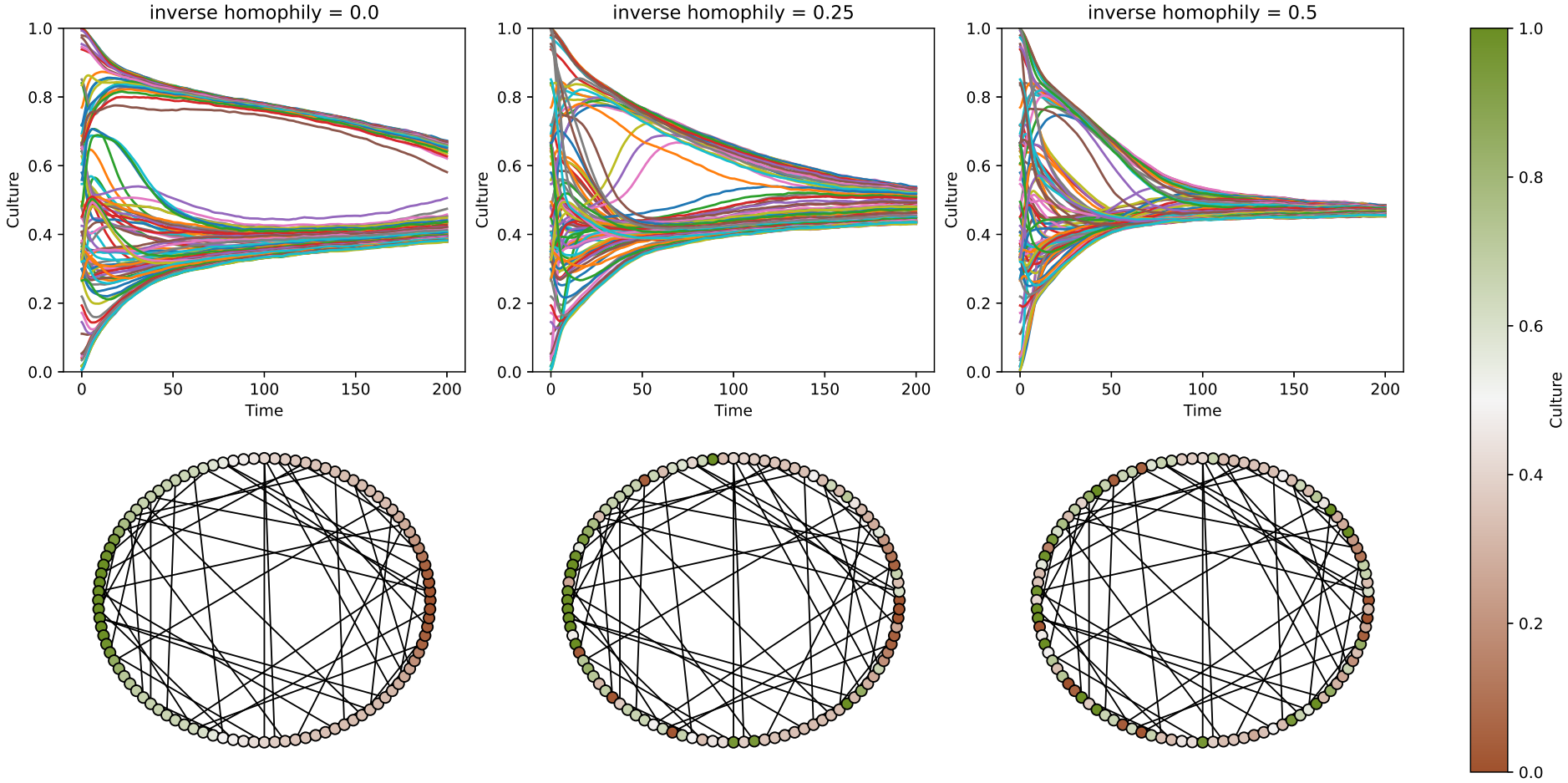
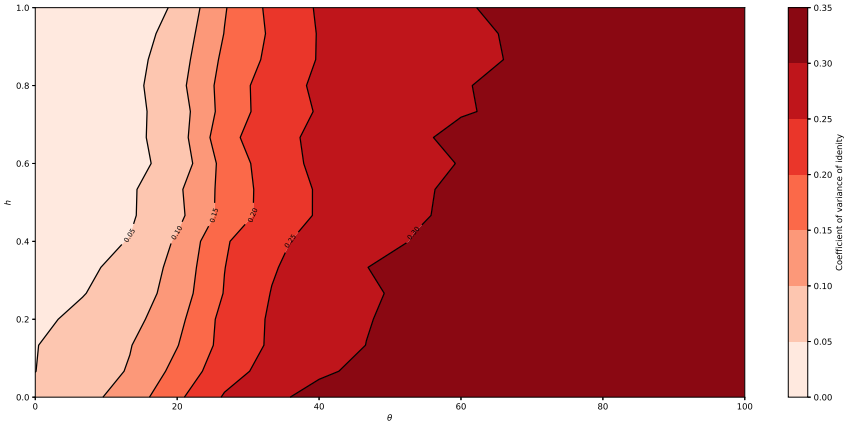


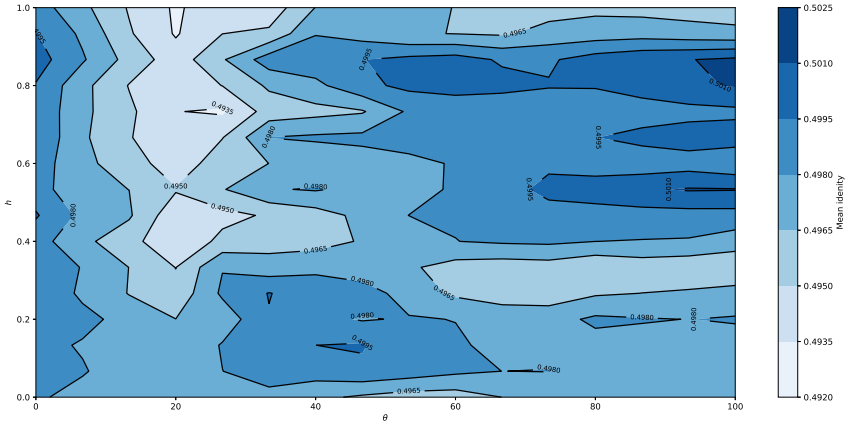
Figure 7:

Decreasing homophily in agent identity leads to more rapid convergence

Results - Varying confirmation bias and identity homophily



(a) Coefficient of variance agent identity



(b) Mean agent identity

Figure 8:

Exploring confirmation bias-homophily phase space

- Increasing confirmation bias leads to greater variance in identity, as agents don´t interact with those different to themselves.
- Importance of confirmation bias in determining the coefficient of variance contrasts emissions

Key takeaways

- **Methodology summary:** Agent-based model of a social network studying behavioural diffusion with identity determining link strength. Adding internal dynamics to culture, through aggregation of multi-dimensional behaviours
- Initial distributions of agent attitudes dominate dynamics
- Changing link strength due to dynamic culture leads to strand formation, delayed consensus formation

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